# OPPONG KWABENA ADUTWUM

## ASPIRING MACHINE LEARNING ENGINEER

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development


## CONTACT

✉ oppongkwabena1777knust@gmail.com
☎ +233 53 529 9655
⌂ Kumasi, Ghana
↗ linkedin.com/in/kwabena-oppong-adutwum
↗ github.com/Oppong-py
↗ leetcode.com/u/Oppong-py
↗ oppong-py.github.io


## TECHNICAL SKILLS

**Programming Languages**
- Python
- C++

**Data & Analytics**
- NumPy
- SQL
- Jupyter Notebooks

**AI & ML Tools**
- Scikit-learn
- Groq AI / Anthropic Claude API

**Web & Backend**
- React Native / Expo

**Dev Tools & Design**
- Git / GitHub
- VS Code
- Figma
- Linux (Kali)

**Core Knowledge**
- Machine Learning Fundamentals
- PID Control / Robotics
- Data Structures & Algorithms
- OSI & TCP/IP Networking

## LANGUAGES

- English — Native
- French — Intermediate
- Twi — Native

## CERTIFICATIONS

- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)

## GITHUB HIGHLIGHTS

- ml-specialization-coursera
  ML study notebooks & practice code
- oppong-py.github.io
  Personal portfolio site

## MEMBERSHIPS

- AmaliTech Coding Club (2025–Present)
- Microsoft User Group Ghana (2025–Present)
- Claude Builder Club — KNUST (2025–Present)
- Prempeh College Robotics Club (2023–2025)

## PROFESSIONAL SUMMARY


First-year Computer Science student at Kwame Nkrumah University of Science and Technology with a strong interest in machine learning, robotics, and embedded systems. National Robotics Champion (Robofest Ghana 2025) and Regional Cybersecurity Champion (2024). Practical project experience includes a PID-controlled autonomous robot, a solar-powered IoT dispenser, and a cross-platform mobile app.

Currently studying Andrew Ng's Machine Learning Specialization (Stanford/Coursera) with focus on supervised learning, vectorization, and NumPy. Working across Python, C++, Scikit-learn, React Native, and Expo through coursework and personal projects.


## EXPERIENCE

### Machine Learning Trainee — Zap Tech Technologies

*Remote | May 2025*

- Completed a 2-week introductory ML program covering data collection, preprocessing, model training, and evaluation.
- Cleaned datasets using Pandas and NumPy — handled missing values and feature scaling.
- Trained classification models with Scikit-learn and compared performance using accuracy, precision, recall, and F1-score.
- Documented experiments and results in Jupyter notebooks.

### Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG)

*Ghana | Oct 2022 – Present*

- Joined as a volunteer in October 2022 for community support projects donating food, sanitary items, and clothing to rural communities.
- Promoted to Online Safety Analyst (August 2023), managing the organisation's social media and website with a focus on internet security.
- Runs speaking engagements during semester breaks with girls from orphanages and rural communities on online safety topics.

### Co-Founder & Software Developer — Arethos Tech Initiatives

*Kumasi, Ghana | 2026 – Present*

- Co-building Arethos, a cross-platform mental wellness app for KNUST students with AI-powered journaling, mood tracking, and reflections (React Native, Expo SDK 54, SQLite, Groq + Anthropic AI dual-fallback). Project in active development.
- Engineered custom haptic feedback (9 iterations), offline-first SQLite storage, and high-fidelity Figma component design systems; coordinating sprint tasks via Git.
- Won 2nd Place at ACES CodeFest 2026 UI/UX Challenge — redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5, building a clickable Figma prototype with real-time collaboration.

## EDUCATION

### Kwame Nkrumah University of Science & Technology (KNUST)

*B.Sc. Computer Science | Kumasi, Ghana | Expected 2029*

**Relevant Coursework:**
- Discrete Mathematics — Logic, set theory, combinatorics, graph theory, proof techniques.
- Probability and Statistics — Probability distributions, Bayesian inference, hypothesis testing, regression analysis.
- Networking & Internet — OSI/TCP/IP models, IPv4 addressing, subnetting, network protocols.
- Calculus I, II & III — Limits, derivatives, integrals, partial and differential equations, hyperbolic and exponential differentiation
- C++ / Computer Science & Mathematics Programming.

**Self-Directed Study:**
- Andrew Ng's Machine Learning Specialization (Stanford / Coursera) — in progress. Covering supervised learning, cost functions, gradient descent, vectorization, and multiple linear regression.
- Independent NumPy deep-dive — arrays, slicing, reshaping, broadcasting, dot products, vectorized operations vs loops, views vs copies.

### Prempeh College (Senior High School)

*WASSCE — General Science | Kumasi, Ghana | Graduated September 2025*

- Core subjects: Core Mathematics, Elective Mathematics, Physics, Chemistry, English Language, Integrated Science, ICT.

**Extracurricular Leadership:**
- Led the Prempeh College Cybersecurity Club (2023–2025). Organised weekly sessions on network security, ethical hacking, and CTF challenges. Mentored 7 junior students.
- Active member of the Robotics Club (2023–2025). Won the Robofest Ghana 2025 National Championship.
- Represented Prempeh College at International Public Policy Forum (2024–2025), organised by Brewer Foundation and New York University (NYU).

## PROJECTS

### Autonomous Line-Following Robot

*LEGO MINDSTORM EV3 · PID Control Algorithm*

- Designed PID control algorithm for autonomous line navigation — National Champion at Robofest Ghana 2025.
- Tuned Kp, Ki, Kd through systematic testing. Achieved sub-second course correction at high speed.
- Optimised sensor-motor feedback loops under strict competition time limits — directly applicable to reinforcement learning and robotics control.

## INTERESTS

- Mathematics
- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

## REFERENCES

- Emmanuel Owusu Ansah
  Senior Teacher, Mathematics — Prempeh College
  +233 54 538 3894
- Kwadwo Gyan-Darkwa
  Physics Teacher — Prempeh College
  +233 244 940 819
- Sally Anokye-Yeboah (Mrs)
  Executive Director — SFLIG
  +233 559 1864 / +233 244 212 953

### Class Attendance Monitoring System

*React Native · Django · PostgreSQL · Bluetooth*

- Architected biometric attendance platform serving 50 students in prototype testing, reducing manual roll-call by ~80%.
- Backend: Django REST Framework with PostgreSQL. RESTful API for registration, attendance logging, analytics.
- Frontend: React Native mobile app for cross-platform deployment.
- Hardware: BLE communication for proximity-based check-in via fingerprint scanner.

### Solar-Powered Automated Water Dispenser

*Arduino · Ultrasonic Sensors · C++ · Solar*

- Built a solar-powered water dispensing system using dual ultrasonic sensors (HC-SR04) to detect container placement and monitor fill level in real time, automatically controlling a solenoid valve.
- Implemented debounce logic to prevent false triggers, buzzer for audio feedback, and LED status indicators for system state.
- Solar power with voltage regulation for off-grid deployment in areas without reliable electricity.

### Arethos — Mental Wellness App (3-Person Team, In Development)

*React Native · Expo SDK 54 · SQLite · Groq & Anthropic AI*

- Co-building a cross-platform mental wellness app for KNUST students featuring AI-powered journaling, mood tracking, and reflections with dual-AI fallback architecture.
- Engineered custom haptic feedback, offline-first SQLite storage, and PanResponder-based gesture handling. Voice recording with real-time waveform visualisation.

## ◆ HACKATHONS & COMPETITIONS

**National Cybersecurity Challenge** — Regional Champion (2024)
- Knowledge-based competition: network security, threat analysis, incident response, cryptography. Represented Prempeh College across the Ashanti Region.

**Robofest Ghana** — National Champion — Robotics (2025)
- Design, build, programme autonomous robots for timed tasks. Won National Championship against teams from across the country.

**ACES CodeFest 2026** — 2nd Place — UI/UX Design (July 2026)
- Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5. Built a high-fidelity clickable prototype in Figma with real-time collaboration, focusing on visual hierarchy, touch-friendly navigation, and user flows.

**International Public Policy Forum (IPPF)** — Competitor (2024–2025)
- Global debate competition by Brewer Foundation & New York University (NYU). Critical analysis, evidence-based argumentation.

## ◆ AWARDS & HONOURS

- National Champion — Robofest Ghana 2025 (Autonomous Robotics)
- Regional Champion — National Cybersecurity Challenge (2024)
- IPPF Competitor — Brewer Foundation & NYU (2024–2025)
- Hedera Certified Foundation (Apr 2026) — among the earliest blockchain-certified students at KNUST.
- Strong academic performance in Mathematics-heavy coursework: Calculus I, Probability & Statistics, Discrete Mathematics, C++ Programming.

## ◆ LEADERSHIP & VOLUNTEERING

**Club Leader** — Prempeh College Cybersecurity Club (2023–2025)
- Vice President of the club. Weekly sessions on networking, ethical hacking, Kali Linux, Wireshark. Mentored 7 junior students.

**Participant** — International Public Policy Forum (IPPF) (2024)
- Global policy debate. Research, argumentation, and collaborative problem-solving with international peers.

**President** — Pearson House Fellowship, Prempeh College (2024–2025)
- Led the Christian fellowship of Pearson House. Mentored 12 aspirants preparing for executive leadership roles.

**Organizing Secretary** — PENSA, Prempeh College (2024–2025)
- Coordinated events and programmes as the main organizing secretary for the campus Christian fellowship.

## ◆ PERSONAL DEVELOPMENT

- Portfolio website: oppong-py.github.io — built with vanilla HTML/CSS/JS, hosted on GitHub Pages.
- Technical blog posts: 'From PID Control to Reinforcement Learning' and 'Three Bugs That Taught Me More Than Any Tutorial'.